

EKALAIVAN SUBRAMANIAN

Embedded Software Developer

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PROFESSIONAL SUMMARY

Embedded Software Developer with 1.5 years of experience building real-time, EtherCAT-based industrial control systems for automotive hydraulic and servo test benches. Skilled in force and displacement control, sensor calibration and I/O integration on Linux. Delivered fully integrated test bench systems for Japanese and Thai customers, from requirements analysis through final deployment.

TECHNICAL SKILLS

Languages: C, C++, Python, Shell Scripting

Protocols: EtherCAT, CAN-FD, MQTT, TCP/IP, UART, I2C, SPI

Operating Systems: RTOS, Embedded Linux (Ubuntu, Debian, LinuxCNC, Xenomai), systemd, Windows

Hardware/Boards: ARM Controllers (STM32F407, STM32F446RE, Raspberry Pi, TI AM243x), RISC Architecture (PIC16F877A), ESP32, ESP8266, IPC (x86), Beckhoff EtherCAT Modules

Peripherals: ADC, DAC, GPIO, PWM, Watchdog

Tools: MATLAB/Simulink, CAN Kvaser, Keil uVision, STM32CubeIDE, MPLAB, proteus, Oscilloscope, Git, Wireshark

Domains: Real-Time Control, Functional Safety (ISO 26262), Linux Networking, OSI Stack, Performance Analysis

Project Methodologies : V-Model SDLC, Scrum, Agile

PROFESSIONAL EXPERIENCE

Embedded Software Developer — ERL Spectra Pvt Ltd, Chennai

March 2025 – Present

- Developed real-time adaptive PID control algorithms for HYDRIVE automotive hydraulic/suspension component testbench systems and performed precise actuator positioning for international customers.
- Built C/C++ application-layer software for force and displacement control with deterministic 1ms timing requirements.
- Integrated Beckhoff EtherCAT I/O modules with Mitsubishi servo drivers via the IgH EtherCAT master on PREEMPT_RT and Xenomai real-time Linux.
- Wrote Linux shell scripts and systemd service units for automated service monitoring, watchdog recovery, and startup sequencing.
- Responsible for end-to-end hardware/software integration (HSI) for complete test bench system, including GUI and application firmware.
- Supported customer demonstrations, acceptance testing for deployments in Japan and Thailand with zero critical post-deployment defects.
- Utilized MATLAB/Simulink and Kvaser (Virtual Channels) in CAN interfaces for CAN, CAN-FD Data validation, and network diagnostics.
- Measured and optimized thread/task scheduling latency and jitter under real-time workloads using custom RT C/C++ benchmarking code, cyclictst, and stress-ng.
- Applied CPU isolation (isolcpus, taskset) to reduce worst-case latency and improve deterministic response for time-critical control loops.
- Good exposure AI models, model training and their tools.

INTERNSHIP

FPGA Design & System Verilog Inter — Object Automation PVT LTD , Remote

Feb 2023 – March 2023

- Supported senior engineers in developing FPGA-based projects, gaining hands-on exposure to core FPGA design workflows and Verilog development practices.
- Key skills & tools used:** Verilog testbench code, digital logic design, SSH, RealVNC Viewer, PowerShell.

PROJECTS

- IPC Platform API for EtherCAT** — Built a custom API abstraction over the open-source IgH EtherCAT master to simplify master-handling for application developers for realtime Control.
- Hydraulic Actuator Testbench Application Development** — Developed the application software for a hydraulic and servo-driven test bench, integrating motor/driver control with sensor feedback for closed-loop actuator testing. Implemented force and displacement control logic to validate actuator performance under real-world load conditions.
- Automatic Fan Speed Control** — Implemented temperature-based automatic fan speed control using the STM32F407 development board and PWM. Integrated analog/digital temperature sensing with timer-generated PWM to drive a 4-pin/2-pin fan, achieving efficient, real-time thermal regulation.
- Smart Irrigation System** — Implemented an automated water irrigation system using the ESP8266 and low-voltage DC motors, with a draft prototype demonstrating sensor-based scheduling and remote control.
- Human Presence Detection System (OpenCV, ESP8266)** — Real-time computer vision pipeline achieving 92% accuracy in variable lighting conditions.
- FPGA Development (Xilinx Vivado)** — Created a basic FPGA design using Xilinx Vivado and K-Layout.

EDUCATION

B.E., Electronics & Communication Engineering — Kongunadu College of Engineering and Technology 2020 – 2024

- CGPA: 7.74/10

HSC — Government Higher Secondary School, Valasiramani 2019 – 2020

- Percentage: 78%

SSLC — Government Higher Secondary School, Valasiramani 2017 – 2018

- Percentage: 88%

CERTIFICATIONS

- Deep Learning Certification — NVIDIA (2024)
- Embedded Systems Professional Training — Pumo Technovation (2024)
- Verilog HDL Programming — Tessolve Semiconductor (2023)

PERSONAL DETAILS

Date of Birth: 28 June 2003

Marital Status: Single

Nationality: Indian

Passport: Valid till August 2033

Gender: Male

Place: Tiruchirapalli, Tamil Nadu

Portfolio: <https://ekalaivans.github.io/>